

Lecture 4: CNN visualization

Plan of the lecture:

1. Classical CNN architectures

- LeNet
- AlexNet
- VGG
- Inception
- ResNet

2. CNN & Visualization

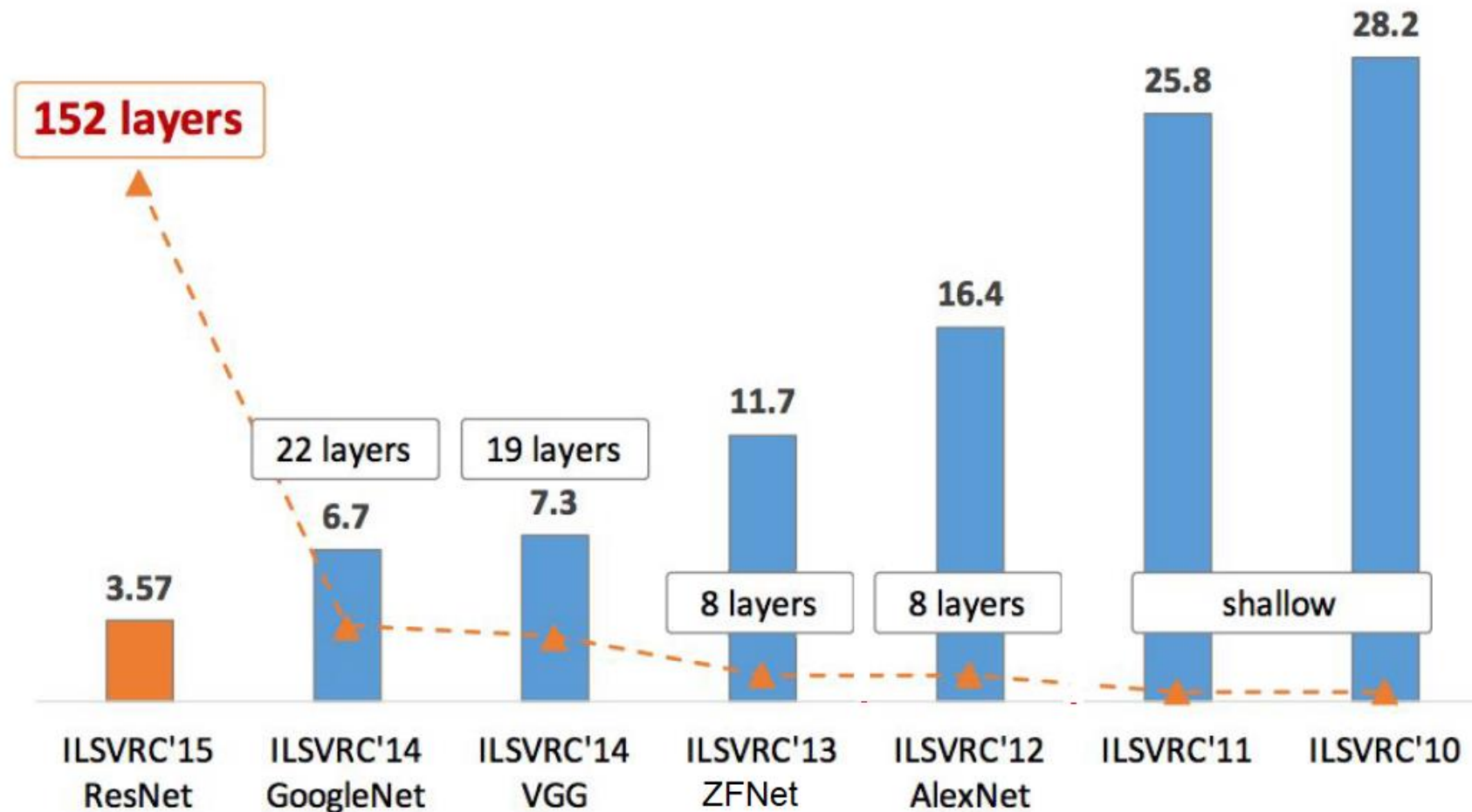
- Visualize filter on first layer
- Visualize filter on last layer
- Saliency Map
 - GradCut
 - Guided backpropagation

□ Most common CNN architectures 5 Case Studies

- - LeNet
- - AlexNet
- - VGG
- - Inception
- - ResNet

Segmentation, Localization, Detection, Visualization

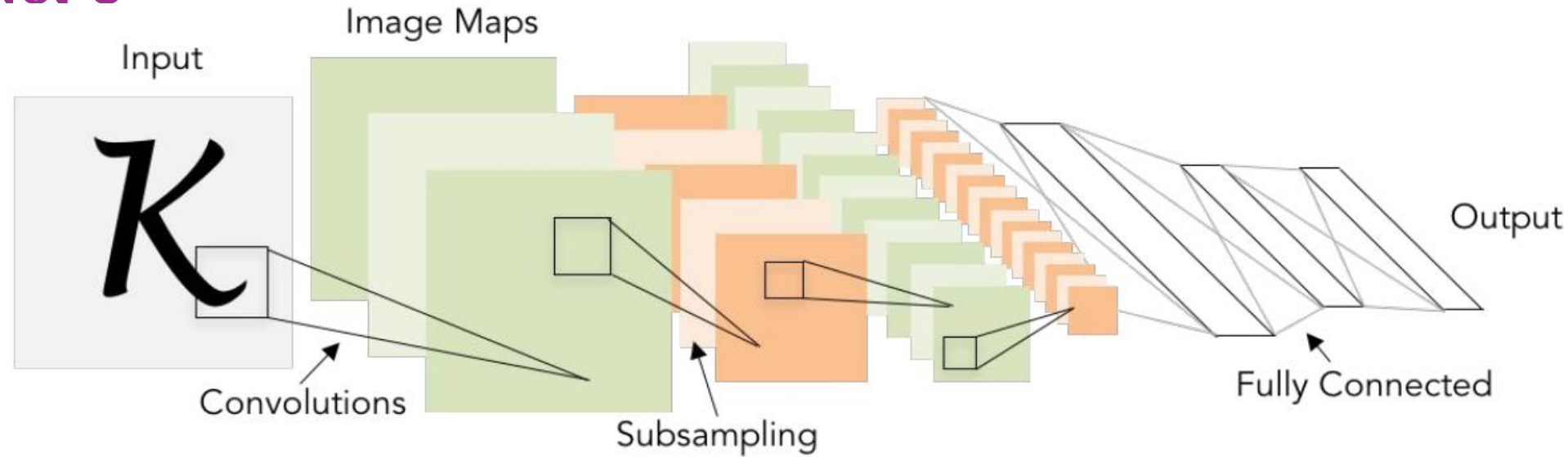
ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



Segmentation, Localization, Detection, Visualization

Case Study: LeNet-5

[LeCun et al., 1998]



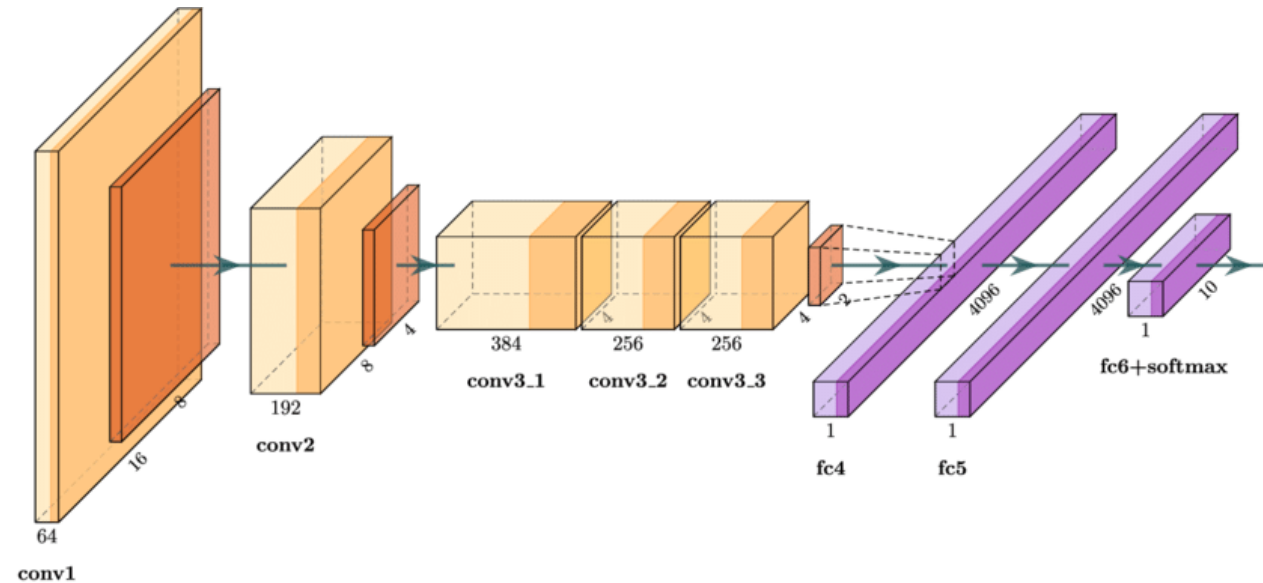
Key contributions

- ❑ **Localized Receptive Fields:** Focus on small regions of the input, reducing computational complexity and improving feature learning.
- ❑ **Weight Sharing:** Each filter shares weights across its receptive field, making the network efficient.
- ❑ **Hierarchical Representation:** Features are extracted in a hierarchical fashion, from simple edges to complex shapes.

Segmentation, Localization, Detection, Visualization

Case Study: AlexNet

[Krizhevsky et al. 2012]



Architecture:

CONV1 MAX POOL1 NORM1

CONV2 MAX POOL2 NORM2

CONV3 CONV4 CONV5

Max POOL3

FC6

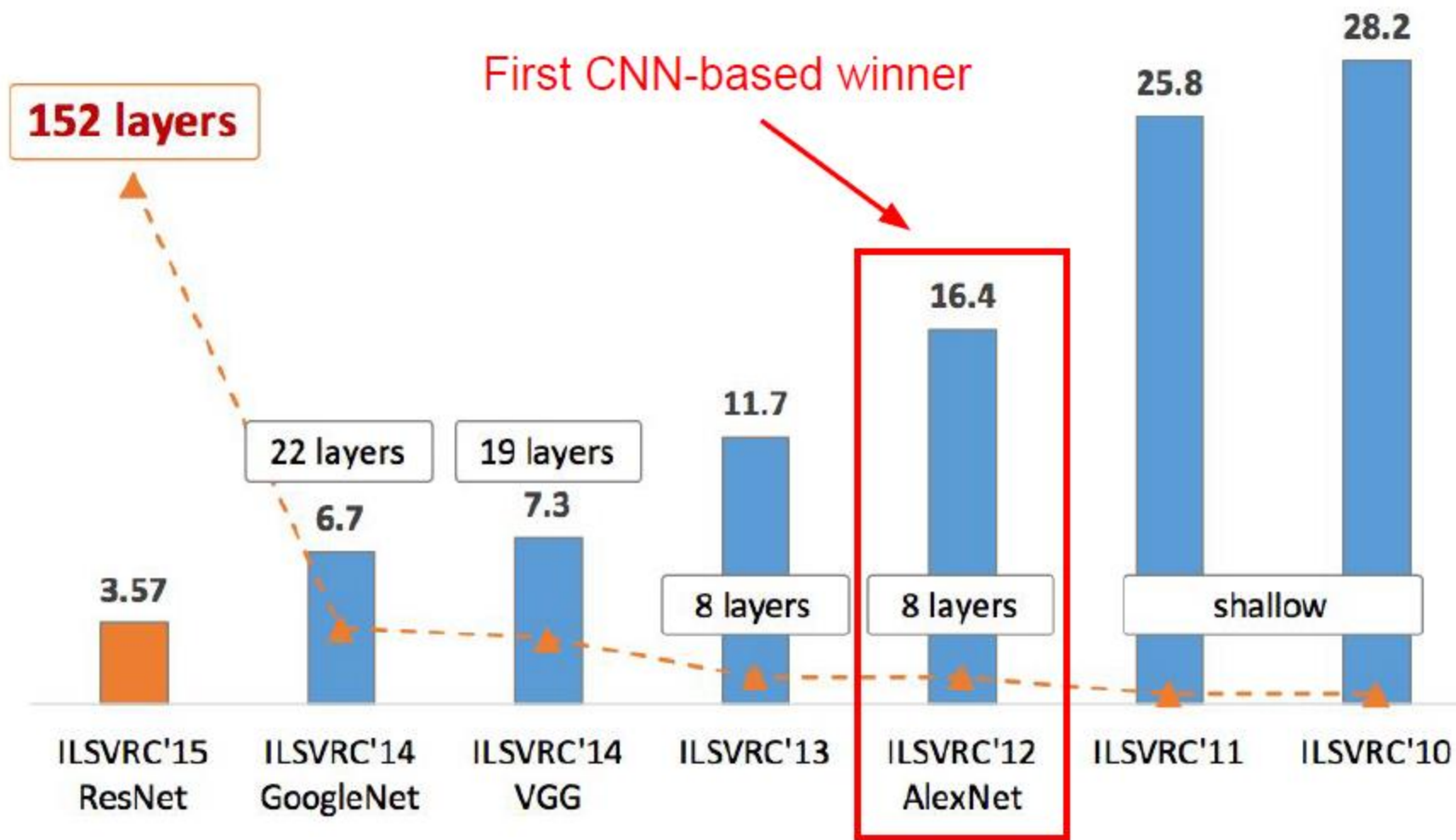
FC7

FC8

Key contributions

- ❑ **Depth:** 8-layer architecture (5 convolutional layers and 3 fully connected layers)
- ❑ **Feature Learning:** It extracted hierarchical features, enabling it to learn complex patterns in images.

Segmentation, Localization, Detection, Visualization



Segmentation, Localization, Detection, Visualization

Case Study: VGGNet

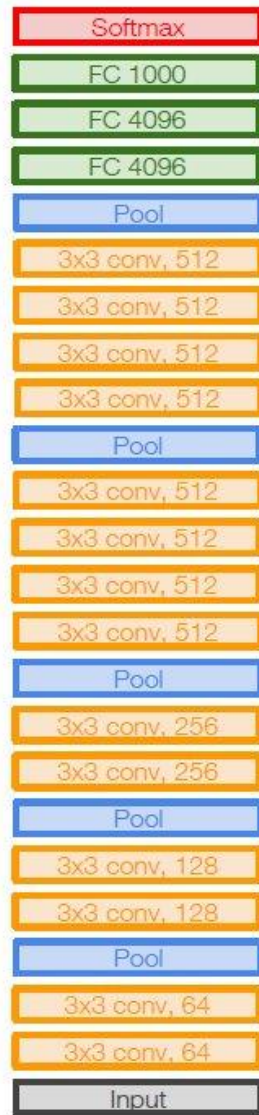
[Simonyan and Zisserman, 2014]

□ Use smaller filters

Remark: stack of three 3x3 conv (stride 1) layers has same effective receptive field as one 7x7 conv layer

□ Deeper means more non-linearities

- **Fully Connected (7 layers)** : output features from the seventh fully connected layer, used as a rich and high-level representation of the input data.
- These features are able to generalize well to other tasks (object detection, image retrieval)
- Extracted and used in transfer learning scenarios



VGG19

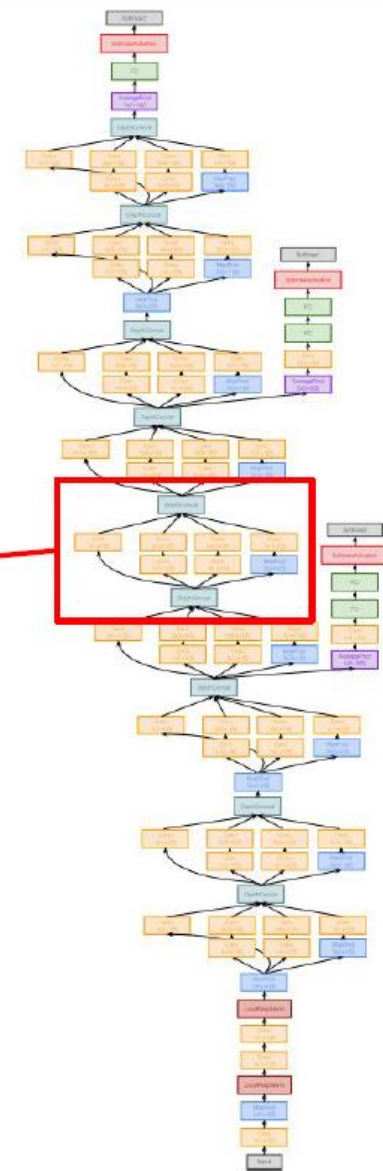
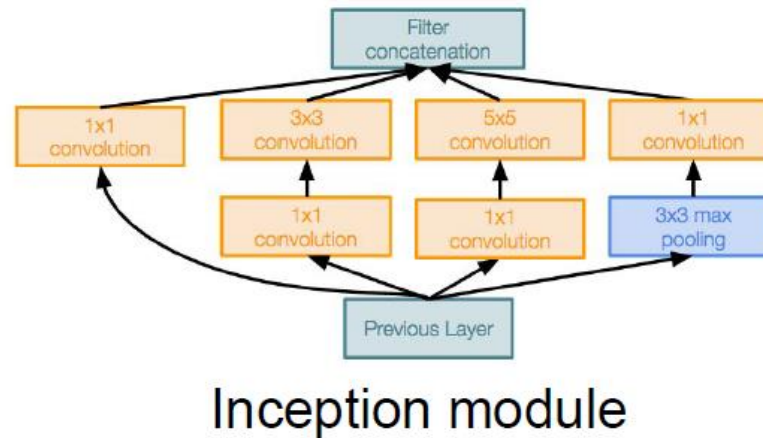
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Case Study: Inception

[Szegedy et al., 2014]

□ The main idea :

Design a good local network topology (network within a network) and then stack these modules on top of each other



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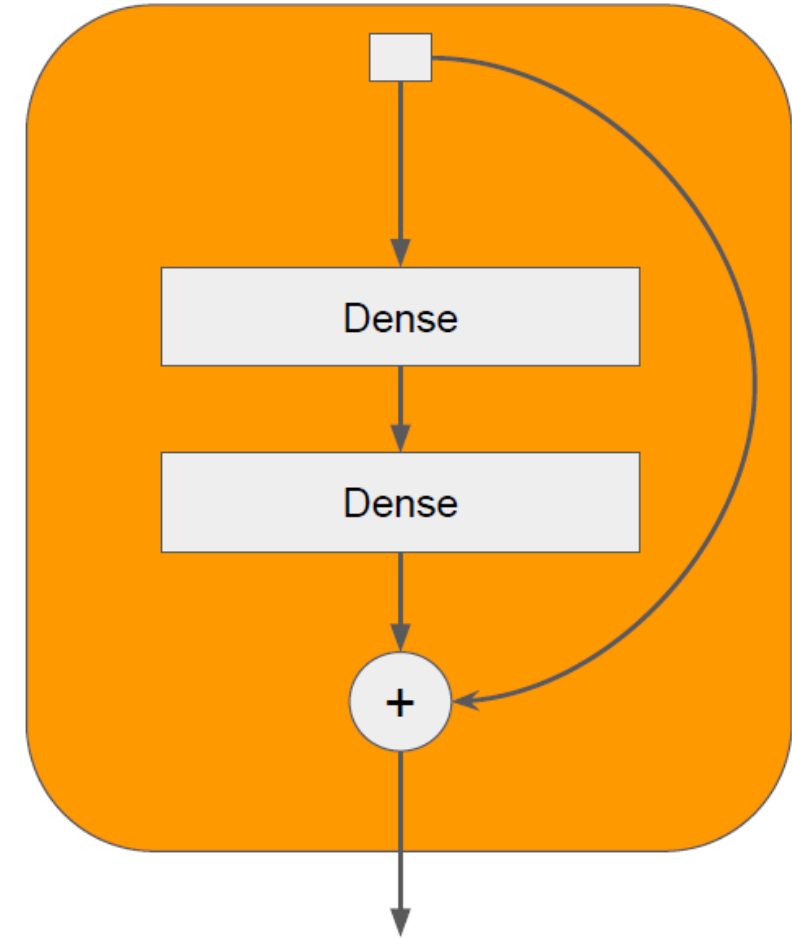
Case Study: ResNet

[He et al., 2015]

- ❑ Suppress vanishing gradient for deeper layers

The idea :

1. Skipp connection by shortcut
2. Evoid perfrmance degradation



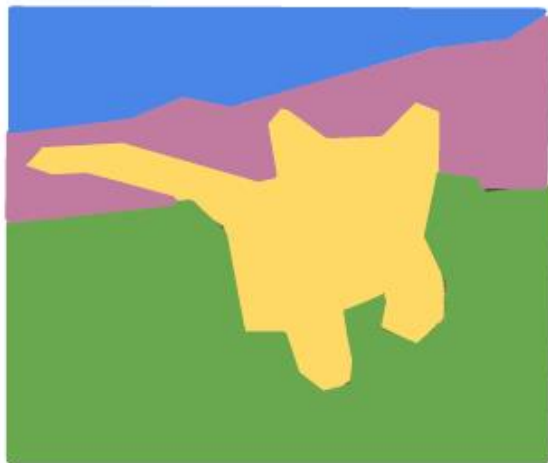
□ Summary: CNN Architectures

- VGG, GoogLeNet, ResNet all in wide use
- ResNet current best default
- Trend towards extremely deep networks
- Significant research centers around design of layer / skip connections and improving gradient flow
- Even more recent trend towards examining necessity of depth vs. width and residual connections

Segmentation, Localization, Detection, Visualization

□ Last Time : computer vision tasks

Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

**Classification
+ Localization**



CAT

Single Object

**Object
Detection**



DOG, DOG, CAT

Multiple Object

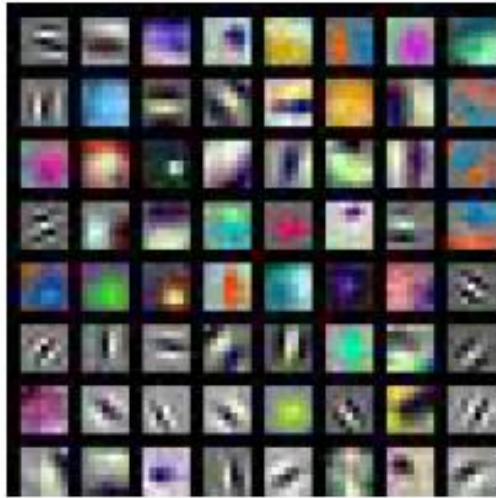
**Instance
Segmentation**



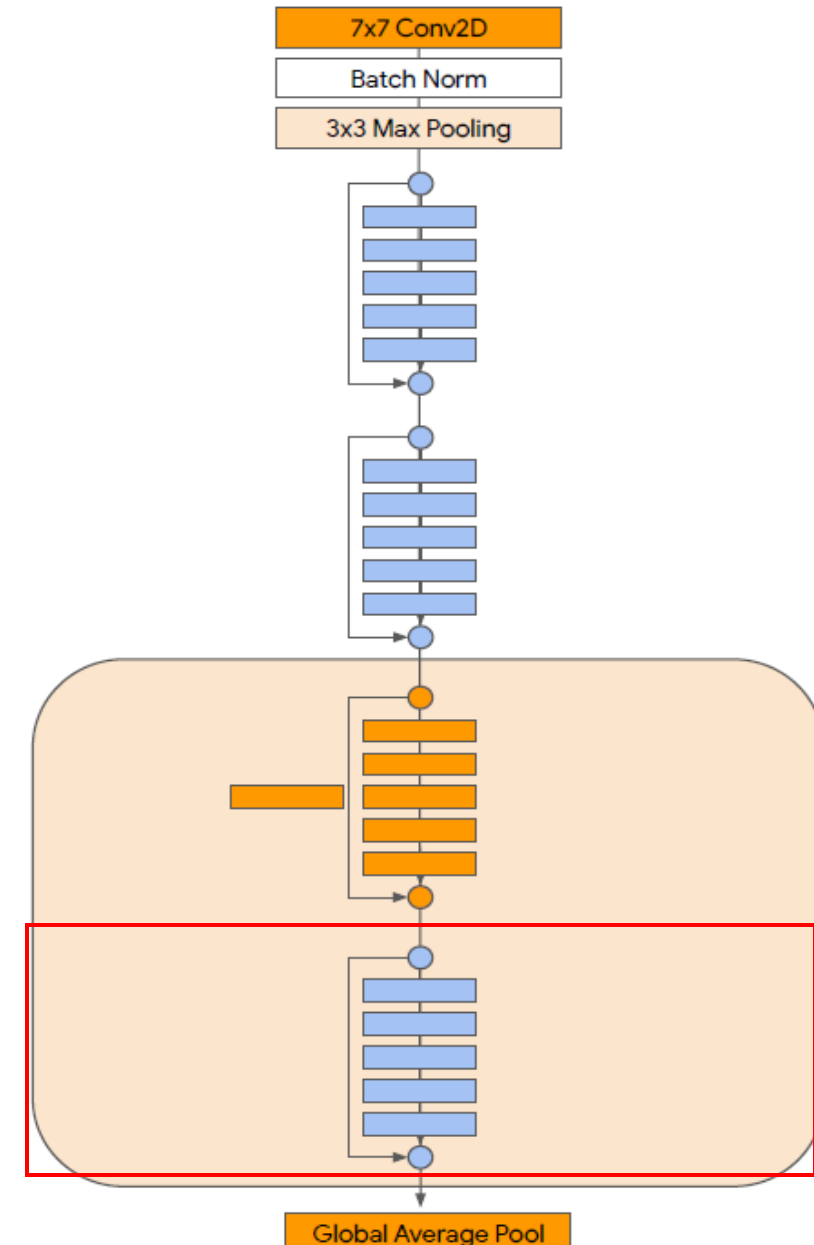
DOG, DOG, CAT

Segmentation, Localization, Detection, Visualization

□ First Layer: Visualize filters



ResNet-18:
64 x 3 x 7 x 7



He et al, "Deep Residual Learning for Image Recognition", CVPR 2016

 Last Layer

-
- The diagram illustrates a 3D CNN architecture for video classification. The input is a 224x224x3 video frame. The architecture consists of several stages: 1. Max pooling (Stride of 4) to 112x112x3. 2. Convolution (5x5) to 55x55x48. 3. Max pooling to 27x27x128. 4. Convolution (3x3) to 27x27x128. 5. Convolution (3x3) to 192x192x128. 6. Convolution (3x3) to 192x192x128. 7. Convolution (3x3) to 192x192x128. 8. Convolution (3x3) to 128x128x13. 9. Convolution (3x3) to 128x128x13. 10. Max pooling to 64x64x128. 11. Global Average Pooling (GAP) to 1x1x128. 12. Dense layer (1000) for classification. A red box highlights the GAP and Dense layers.

29/11/24 K.Kadri DVRC Computer Vision Lecture 4

Segmentation, Localization, Detection, Visualization

□ Saliency Maps

➤ How to tell which pixels matter for classification

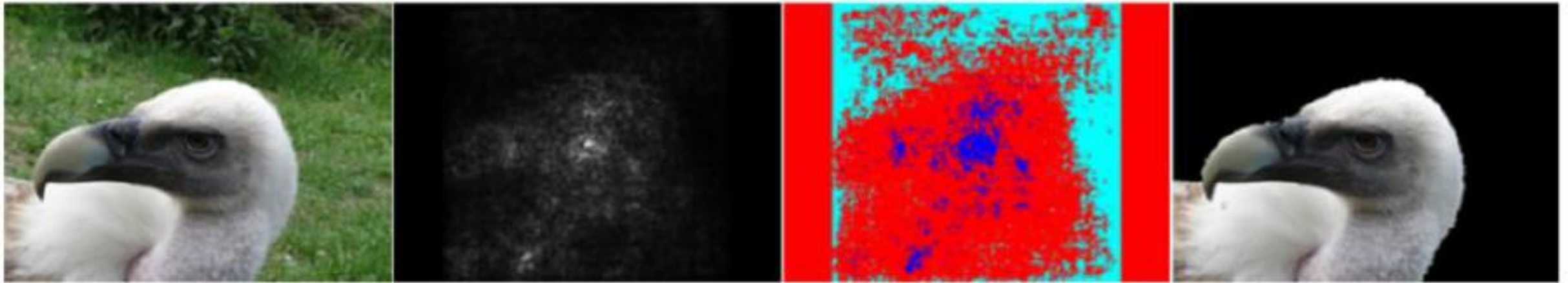
1. Compute gradient of (unnormalized)
2. Class score with respect to image pixels
3. Take absolute value and max over RGB channels



Segmentation, Localization, Detection, Visualization

□ Saliency Maps: Segmentation without supervision

➤ How to tell which pixels matter for classification



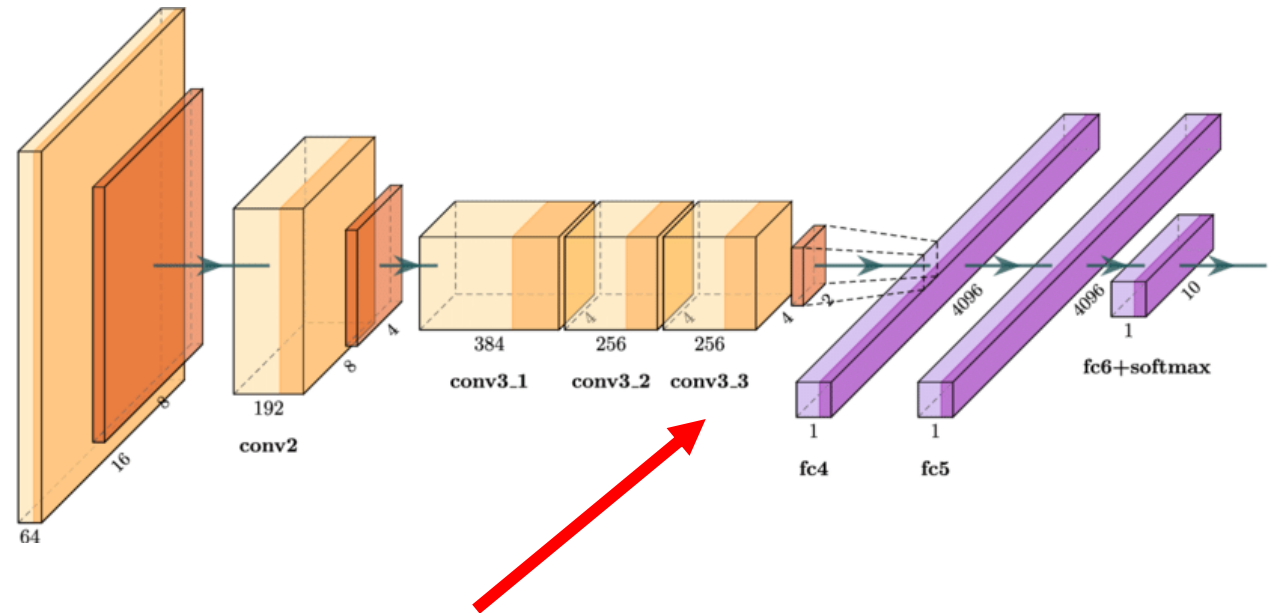
- Use GrabCut on saliency map
- Background suppression

Simonyan, Vedaldi, and Zisserman, “Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps”, ICLR Workshop 2014.

Segmentation, Localization, Detection, Visualization

Intermediate Features via guided backpropagation

➤ How to tell which pixels matter



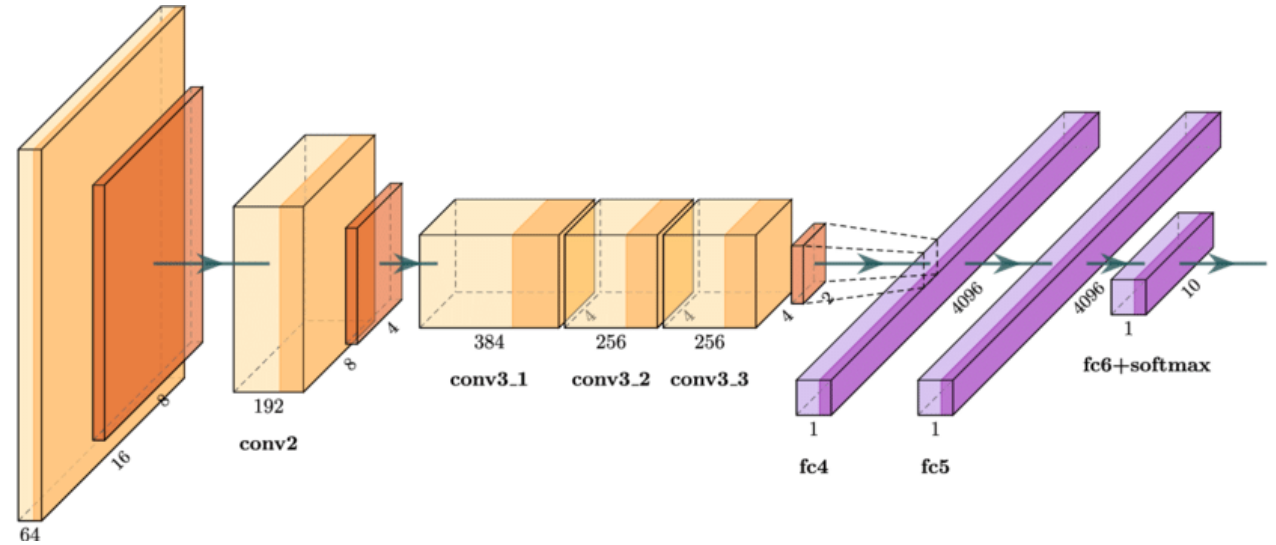
1. Pick a single intermediate neuron, e.g. one value in $128 \times 13 \times 13$ conv3 feature map
2. Compute gradient of neuron value with respect to image pixels

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Intermediate Features via guided backpropagation

➤ How to tell which pixels matter

□ How it works



1. guided backpropagation can tell us which pixels in the input image matter for the correct classification
2. **Guided** backpropagation because we choose('guide') which neurons are to be activated for backpropagation

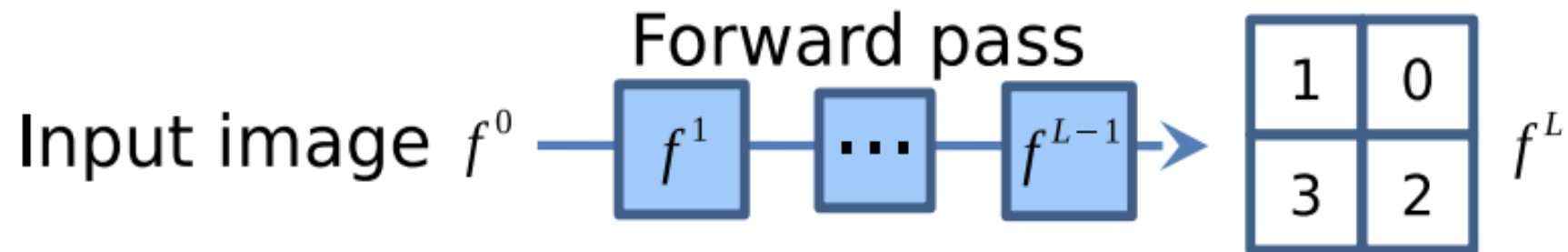
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Intermediate Features via guided backpropagation

How it works

- Guided backpropagation is a gradient-based visualization technique.

1. firstly we do a forward pass till the layer that we are interested in



Consider the feature maps ($f_1, f_2, f_3, \dots, f_L$)

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Intermediate Features via guided backpropagation

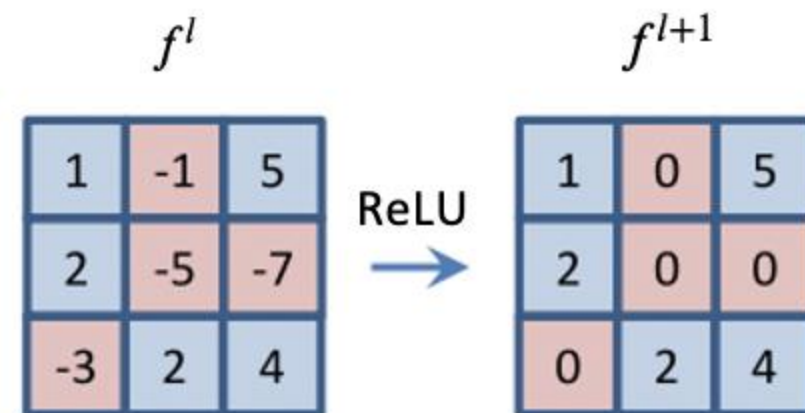
How it works

- Guided backpropagation is a gradient-based visualization technique.

2. The input values in the previous layer are negative, the output after the ReLU operation are set to 0

$$f_i^{l+1} = \text{ReLU}(f_i^l) = \max(f_i^l, 0)$$

Forward pass



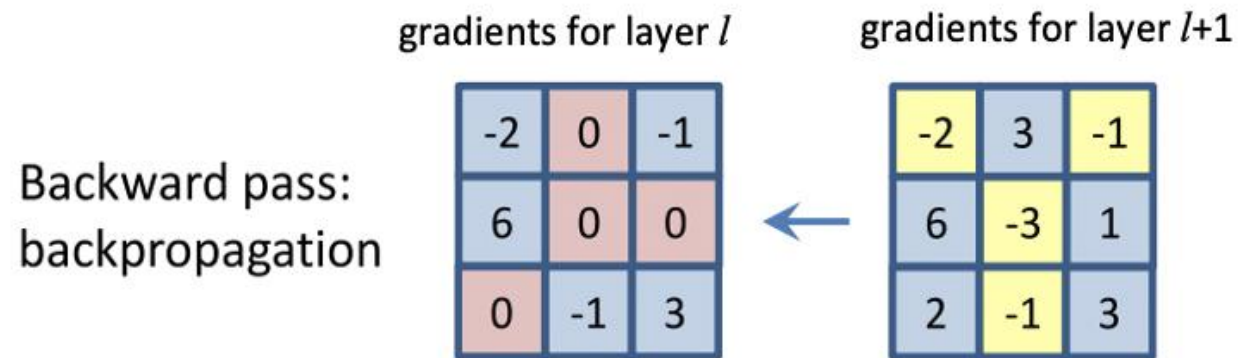
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Intermediate Features via guided backpropagation

How it works

- Guided backpropagation is a gradient-based visualization technique.

3. For the ReLU, when we do a **traditional** backpropagation, it passes gradient to a previous layer only if the **original input** was positive (the neurons are not dead)



The neurons marked **red** are dead, and the gradients can not be backpropagated.

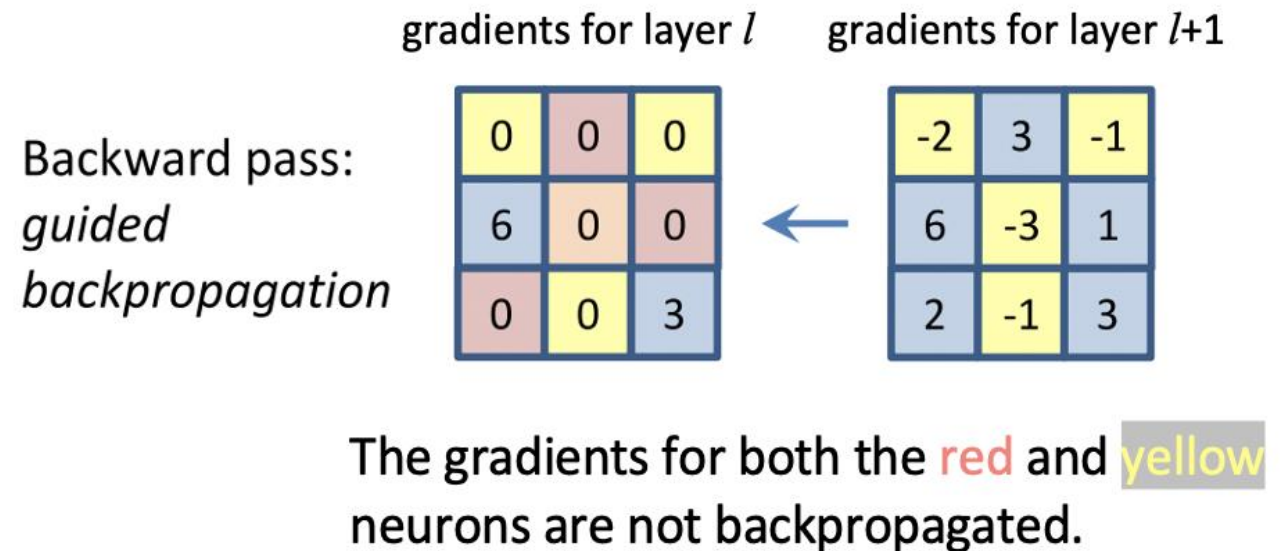
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Intermediate Features via guided backpropagation

How it works

- Guided backpropagation is a gradient-based visualization technique.

4. In the case of **guided** backpropagation, we also do not backpass the negative gradients (the neurons marked yellow in the above picture) to the previous layer



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Intermediate Features via guided backpropagation

How it works

- Guided backpropagation is a gradient-based visualization technique.

5. Given the feature map ($f_1, \dots, f_L$), we are only interested in the neuron at the right bottom corner, which has the gradient value

- We can retain this neuron and set all other neurons to 0.
- Then the gradient for the chosen neuron is backpropagated all the way to the inputs to get a reconstruction

